

Multirate Stein Variational Gradient Descent for Efficient Bayesian Sampling

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Abstract

Many particle-based Bayesian inference methods use a single global step size for all parts of the update. In Stein variational gradient descent (SVGD), however, each update combines two qualitatively different effects: attraction toward high-posterior regions and repulsion that preserves particle diversity. These effects can evolve at different rates, especially in high-dimensional, anisotropic, or hierarchical posteriors, so one step size can be unstable in some regions and inefficient in others. We derive a multirate version of SVGD that updates these components on different time scales. The framework yields practical algorithms, including a symmetric split method, a fixed multirate method (MR-SVGD), and an adaptive multirate method (Adapt-MR-SVGD) with local error control. We evaluate the methods in a broad and rigorous benchmark suite covering six problem families: a 50D Gaussian target, multiple 2D synthetic targets, UCI Bayesian logistic regression, multimodal Gaussian mixtures, Bayesian neural networks, and large-scale hierarchical logistic regression. Evaluation includes posterior-matching metrics, predictive performance, calibration quality, mixing, and explicit computational cost accounting. Across these six benchmark families, multirate SVGD variants

improve robustness and quality-cost tradeoffs relative to vanilla SVGD. The strongest gains appear on stiff hierarchical, strongly anisotropic, and multimodal targets, where adaptive multirate SVGD is usually the strongest variant and fixed multirate SVGD provides a simpler robust alternative at lower cost.

Keywords: Stein variational gradient descent, Bayesian computation, multirate integration, particle-based variational inference, adaptive time stepping

1. Introduction

Modern Bayesian inference often requires efficient sampling from complex, high-dimensional, and multimodal posterior distributions. As these problems grow in scale and complexity, traditional sampling methods face significant computational and statistical challenges. The advent of automatic differentiation has enabled the widespread use of gradient-based samplers, making methods such as stochastic gradient Langevin dynamics (SGLD) and stochastic gradient Hamiltonian Monte Carlo (SGHMC) practical and popular choices for large-scale applications [1, 2]. While these approaches advance a single Markov chain using noisy gradients, an alternative paradigm is offered by interacting particle methods, which evolve a population of particles whose empirical distribution approximates the target posterior. This population-based view is central to particle-based variational inference [3]. Within this class, SVGD is a prominent deterministic representative, and recent work has examined acceleration, particle degeneracy, and kernel design in related Stein-particle methods [3, 4, 5]. It opens new avenues for improv-

17 ing sampling efficiency and robustness, especially in challenging inference
18 settings.

19 A useful unifying perspective is to view interacting particle methods as
20 discretizations of a particle flow (possibly with a stochastic component) that
21 transports an initial ensemble toward the target distribution [6]. Here, we
22 focus on Stein variational gradient descent (SVGD) [7] as a representative in-
23 teracting particle method. SVGD constructs a velocity field for the particles
24 based on the target density and a reproducing kernel Hilbert space (RKHS)
25 structure, and updates the particles by discretizing this flow [7]. The veloc-
26 ity field naturally separates into an attractive drift, which moves particles
27 toward high-density regions, and a repulsive interaction, which maintains
28 particle diversity and support coverage of multiple modes.

29 This drift-repulsion structure creates the main numerical challenge stud-
30 ied in this paper. When particles crowd, repulsion can become stiff and
31 destabilize updates. On the other hand, the drift can require finer resolution on
32 anisotropic targets to follow the geometry of $\log p$ closely. A single step size
33 must therefore balance competing requirements and can either over-damp
34 repulsion, leading to loss of diversity and mode coverage, or under-resolve
35 drift, leading to slow progress.

36 Moreover, repulsion typically requires kernel evaluation at each step, so
37 increasing the stability by reducing the step-size can become computationally
38 expensive. Finally, repulsive forces can weaken in high dimensions or under
39 limited particle resolution, leading to particle collapse and poor support of
40 multiple modes [4, 8]. In practice, the resulting single-step-size compromise
41 is both statistically and computationally inefficient: it can degrade support

coverage and predictive quality while still spending substantial kernel work to stabilize the wrong part of the update.

A common approach to solving systems with significantly different components is to use partitioned methods such as implicit-explicit (IMEX) [9, 10] and multirate integration methods [11]. In the context of ordinary differential equations (ODEs), multirate methods exploit a natural splitting of the dynamics into fast and slow components, allowing each to be integrated with a step size appropriate to its scale [12, 13, 14]. For example, in a split ODE of the form $\dot{x} = \phi_{\text{fast}}(x) + \phi_{\text{slow}}(x)$, multirate schemes apply more frequent, smaller steps to the stiff or rapidly varying part, while updating the slow part less often. This approach improves both stability and efficiency in multi-scale dynamical systems, and has been successfully applied in a variety of scientific computing contexts [10, 15].

Motivated by advances in multirate time-integration for multi-scale dynamical systems—including MrGARK, IMEX, and related splitting schemes [12, 13, 14] we propose to integrate the drift and repulsion terms on separate time scales. This leads to a number of multirate SVGD variants, including symmetric splitting, fixed repulsion substepping, and adaptive error-controlled drift substepping.

To assess the effectiveness of these methods, we conduct a comprehensive empirical evaluation. We compare our multirate SVGD variants with established approaches from the literature across a range of benchmark problems, using a variety of task-appropriate metrics. Results highlight the strengths and limitations of each method under diverse geometric and statistical challenges.

The main contributions of this work are:

- Multirate SVGD formulations that explicitly separate drift and repulsion and support symmetric splitting and fixed multirate schedules.
- An adaptive error-controlled multirate SVGD variant that selects substeps based on a local error estimator while keeping repulsion stable.
- A broad empirical study showing that multirate SVGD improves robustness and quality-cost tradeoffs compared to SVGD and representative chain methods, with the strongest gains on anisotropic, multimodal, and hierarchical targets.
- A benchmark suite spanning many different targets, equipped with appropriate diagnostics, early stopping, performance evaluation metrics for efficiency comparisons.

We present our work as follows. Section 2 presents the SVGD variants and the multirate construction. Section 3 describes the benchmarks, diagnostics, and experimental protocol. Section 4 reports the results. Section 5 summarizes the findings and discusses directions for future work.

2. Method

2.1. Background: SVGD as a particle flow

Let $p(x)$ denote a target density on $\mathbb{R}^d$ and let $\{x_i^t\}_{i=1}^N$ be a set of particles at iteration t . SVGD evolves particles by following a deterministic interacting-particle flow whose velocity field is constrained to RKHS [7]. A

88 first-order discretization of this flow is the update

$$x_i^{t+1} = x_i^t + h \phi(x_i^t), \quad (1)$$

89 where $h > 0$ is a step size and the SVGD velocity field is

$$\phi(x) = \frac{1}{N} \sum_{j=1}^N \left[k(x_j, x) \nabla_{x_j} \log p(x_j) + \nabla_{x_j} k(x_j, x) \right]. \quad (2)$$

90 The first term in eq. (2) is an *attractive* drift that pushes particles toward
 91 high-density regions via the score $\nabla \log p$. The second term is a *repulsive*
 92 interaction that discourages particle collapse. The multirate constructions
 93 below are kernel-agnostic: they apply once a kernel-induced attractive field
 94 and repulsive field have been specified.

95 2.2. Drift–repulsion splitting and multirate integration

96 The central numerical observation motivating this work is that the drift
 97 and repulsion components of eq. (2) can impose different stability constraints
 98 and can naturally evolve on different time scales. We therefore write the
 99 SVGD flow as a split system

$$\dot{x} = f_{\text{rep}}(x) + f_{\text{drift}}(x), \quad (3)$$

100 where, for particle i ,

$$f_{\text{drift}}(x_i) := \frac{1}{N} \sum_{j=1}^N k(x_j, x_i) \nabla_{x_j} \log p(x_j), \quad (4)$$

$$f_{\text{rep}}(x_i) := \frac{1}{N} \sum_{j=1}^N \nabla_{x_j} k(x_j, x_i). \quad (5)$$

101 We next define multirate discretizations that apply different substepping
 102 strategies to f_{drift} and f_{rep} while keeping a common macro-step size h .

103 *Strang-split SVGD*. Define the forward Euler update that advance the split
 104 system by a step size τ :

$$\Psi_{\text{rep}}^{\tau}(x) = x + \tau f_{\text{rep}}(x), \quad \Psi_{\text{drift}}^{\tau}(x) = x + \tau f_{\text{drift}}(x).$$

105 *Strang splitting* [16] is a classical second-order method that constructs a
 106 macro-step of size h by composing such substeps. In our setting, one macro
 107 step takes the form

$$x^{t+1} = \Psi_{\text{rep}}^{h/2} \circ \Psi_{\text{drift}}^h \circ \Psi_{\text{rep}}^{h/2}(x^t), \quad (6)$$

108 which alternates half a repulsion step, a full drift step, and another half
 109 repulsion step. In practice, the half-steps can be implemented as $\frac{h}{2m}$ smaller
 110 substeps for stability.

111 *Fixed multirate SVGD (MR-SVGd)*. Fixed multirate integrators provide a
 112 principled way to allocate different step sizes to fast and slow components of
 113 a split system [11, 12, 13]. In our SVGD setting, however, higher-order multi-
 114 rate schemes are often not cost-effective: their additional stages translate into
 115 multiple evaluations of the drift and repulsion fields per macro step, and each
 116 such evaluation is expensive due to gradient computations and, in particular,
 117 $O(N^2)$ kernel interactions. Moreover, practical accuracy is also limited by
 118 finite-particle effects and by choices such as bandwidth selection and early
 119 stopping, which can mute the benefits of increasing time-integration order
 120 at a fixed computational budget. We therefore utilize a first-order multirate
 121 construction using Euler updates that preserves a macro step of size h for
 122 the drift while resolving the repulsion with m substeps:

$$x^{t+1} = \Psi_{\text{drift}}^h \circ \left(\Psi_{\text{rep}}^{h/m} \right)^m (x^t). \quad (7)$$

Algorithm 1 Fixed multirate SVGD (MR-SVGD)

Require: Particles $\{x_i\}_{i=1}^N$, step size h , repulsion substeps m , kernel k

```
1:  $x \leftarrow \{x_i\}$ 
2: for  $s = 1$  to  $m$  do
3:   compute repulsion  $f_{\text{rep}}(x)$  using eq. (5)
4:    $x \leftarrow x + (h/m) f_{\text{rep}}(x)$ 
5: end for
6: compute drift  $f_{\text{drift}}(x)$  using eq. (4)
7:  $x \leftarrow x + h f_{\text{drift}}(x)$ 
8: return  $x$ 
```

123 This retains a coarse drift step while stabilizing the repulsive interaction
124 when particles crowd, at the cost of additional kernel evaluations. The cor-
125 responding implementation-level procedure is summarized in algorithm 1.

126 *Adaptive multirate SVGD (Adapt-MR-SVGD).* Although fixed- m schedules
127 are simple and robust, they impose a uniform drift resolution along the entire
128 trajectory. As a result, they may spend unnecessary computational effort in
129 benign regions while providing insufficient resolution in locally stiff regimes.
130 To address this imbalance, we choose the number of drift microsteps m adap-
131 tively at each macro step, while retaining one repulsion update per macro
132 step. This follows the standard local-error-control perspective used in ODE
133 integrators, where local error estimates are used to adjust resolution [17, Ch.
134 II]. After the repulsion update, denote the state by $x = \Psi_{\text{rep}}^h(x^t)$. We esti-
135 mate the local drift-discretization error by comparing one drift step of size h

136 with two consecutive drift half-steps of size $h/2$,

$$x_{\text{full}} = x + h f_{\text{drift}}(x), \quad (8)$$

$$x_{\text{half}} = x + \frac{h}{2} f_{\text{drift}}(x), \quad (9)$$

$$\tilde{x}_{\text{full}} = x_{\text{half}} + \frac{h}{2} f_{\text{drift}}(x_{\text{half}}). \quad (10)$$

137 We then form the relative discrepancy

$$\rho = \frac{\|\tilde{x}_{\text{full}} - x_{\text{full}}\|_2}{\|x\|_2 + \epsilon}, \quad (11)$$

138 which serves as a dimensionless local error indicator for the drift update.
 139 $\epsilon > 0$ is a small constant to avoid division by zero. For a first-order drift
 140 integrator, this local error behaves as [17, Ch. II.4]:

$$\rho(h) = Ch^2 + \mathcal{O}(h^3),$$

141 where C is the principal error constant dependent on f and its derivatives.
 142 If the drift step is replaced by m microsteps of size h/m , then

$$\rho_m \approx C(h/m)^2 \approx \rho/m^2.$$

143 Imposing the target local error tolerance $\rho_m \approx \text{Tol}$ gives $m \approx \sqrt{\rho/\text{Tol}}$. We
 144 therefore use the clipped integer controller

$$m \leftarrow \text{clip}\left(\left\lceil \sqrt{\rho/\text{Tol}} \right\rceil, m_{\min}, m_{\max}\right), \quad (12)$$

145 which increases m when $\rho > \text{Tol}$ and decreases it when $\rho \leq \text{Tol}$, while
 146 enforcing the robustness bounds $m_{\min} \leq m \leq m_{\max}$. The macro-step update
 147 is then

$$x^{t+1} = \left(\Psi_{\text{drift}}^{h/m}\right)^m \circ \Psi_{\text{rep}}^h(x^t), \quad m = m(x^t, h). \quad (13)$$

148 This strategy increases drift resolution only when the local error indicator
 149 suggests stiffness or rapid variation in the drift component. Algorithm 2
 150 provides the complete algorithm for this method.

151 *Remark 1.* In practice, if $m \leq 2$ we can reuse the already computed, locally-
 152 higher-order solution $\tilde{x}_{\text{full}}$ to avoid extra drift evaluations.

153 3. Experimental Setup

154 We evaluate multirate SVGD variants on six benchmark groups: a 50D
 155 Gaussian target, a suite of 2D synthetic targets, UCI logistic regression, a
 156 2D Gaussian mixture (mix8), a one-hidden-layer Bayesian neural network
 157 (BNN), and a large-scale hierarchical logistic-regression (HLR) benchmark.
 158 The comparison focuses on both statistical quality and computational cost.
 159 This section defines the shared diagnostics and protocol.

160 3.1. Evaluation Metrics

161 Evaluation combines distributional fidelity, predictive performance, mix-
 162 ing, and explicit computational cost accounting. Metrics used across multiple
 163 benchmarks are defined here once. Benchmark-specific metrics are intro-
 164 duced in their respective sections.

165 *Moment fidelity.* When reference moments are available, we report $e_\mu =$
 166 $\|\hat{\mu} - \mu_{\text{ref}}\|_2$ and $e_\Sigma = \|\hat{\Sigma} - \Sigma_{\text{ref}}\|_F$, where $\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i$ and $\hat{\Sigma} = \frac{1}{N} \sum_{i=1}^N (x_i -$
 167 $\hat{\mu})(x_i - \hat{\mu})^\top$. These summarize first- and second-order bias, which is critical
 168 in the Gaussian and synthetic-target settings.

169 *Stein discrepancy.* For distributional fidelity, we use kernel Stein discrepancy
 170 (KSD) with score $s_p(x) = \nabla_x \log p(x)$ and an RBF base kernel $k_h(\cdot, \cdot)$ [18, 19].
 171 The corresponding Stein kernel is

$$\begin{aligned} \kappa_p(x, x') &= s_p(x)^\top s_p(x') k_h(x, x') + s_p(x)^\top \nabla_{x'} k_h(x, x') \\ &\quad + s_p(x')^\top \nabla_x k_h(x, x') + \text{tr}(\nabla_x \nabla_{x'} k_h(x, x')) , \end{aligned} \quad (14)$$

172 and the empirical discrepancy is

$$\text{KSD}(X; p) = \left(\frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \kappa_p(x_i, x_j) \right)^{1/2} . \quad (15)$$

173 KSD measures how far the empirical particle distribution is from satisfying
 174 the target’s Stein identity. Intuitively, it acts as a global goodness-of-fit
 175 metric: it is small when the particles collectively match the target law, and
 176 large when they exhibit systematic errors such as bias, incorrect spread, or
 177 missed modes.

178 *Mean log-density fit.* We complement KSD with the empirical mean log-
 179 density

$$\overline{\log p}(X) = \frac{1}{N} \sum_{i=1}^N \log p(x_i), \quad (16)$$

180 which measures how strongly the particles concentrate in high-density regions
 181 of the target. Unlike KSD, this is a pointwise criterion: it rewards samples
 182 for lying in locally plausible regions, but by itself it does not certify global
 183 distributional agreement. Reporting eqs. (15) and (16) therefore separates
 184 global distributional fidelity from local density concentration.

185 *Predictive performance and calibration.* For the binary-classification exper-
 186 iments (UCI and BNN tasks), we report accuracy, negative log-likelihood,

187 and expected calibration error ¹,

$$\text{NLL} = -\frac{1}{M} \sum_{m=1}^M [y_m \log \hat{p}_m + (1 - y_m) \log(1 - \hat{p}_m)], \quad (17)$$

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{B}_b|}{M} |\text{acc}(\mathcal{B}_b) - \text{conf}(\mathcal{B}_b)|. \quad (18)$$

188 In all predictive experiments, ECE is computed with $B = 10$ equal-width
 189 bins on $[0, 1]$. Together, these metrics capture thresholded classification per-
 190 formance, probabilistic predictive fit, and calibration quality.

191 *Mixing efficiency.* To quantify dependence, we report ESS as a supplemen-
 192 tal univariate mixing diagnostic, computed from the first coordinate using
 193 a standard autocorrelation-based estimator with initial-positive truncation
 194 [20]. For particle methods at a checkpoint, the diagnostic sequence is formed
 195 from the first coordinate across particles; for single-chain baselines, it is
 196 formed from a rolling window of recent iterates. Because this quantity is
 197 one-dimensional, we do not interpret ESS as a full multivariate efficiency
 198 measure but as a complementary statistic.

199 *Cost accounting.* Alongside wall-clock time, we track gradient and kernel
 200 evaluation counts to support hardware-agnostic efficiency comparisons across
 201 methods with different workloads.

202 *Remark 2.* For the common metrics above, lower is better for e_μ , e_Σ , KSD,
 203 NLL, ECE, wall-clock time, and gradient- and kernel-evaluation counts, whereas

¹Accuracy as the fraction of correctly classified examples under a 0.5 threshold on the posterior predictive probability. $\text{acc}(\mathcal{B}_b)$ is the empirical accuracy in bin $\mathcal{B}_b$ under the same threshold, and $\text{conf}(\mathcal{B}_b)$ is the mean predictive probability in that bin.

204 higher is better for ESS, accuracy, and mean log-probability.

205 3.2. Experimental Protocol

206 Across benchmarks, we compare the same method family whenever ap-
 207 plicable: vanilla SVGD, Strang-split SVGD, fixed multirate SVGD (MR-
 208 SVGD), adaptive multirate SVGD (Adapt-MR-SVGD), and the single-chain
 209 baselines SGLD and an SGHMC-style method. The SVGD-family updates
 210 are those defined in section 2, namely eqs. (1), (6), (7) and (13).

211 Single-chain baselines use the standard stochastic-gradient update

$$x_{t+1} = x_t + \eta \nabla \log p(x_t) + \sqrt{2\eta} \xi_t, \quad \xi_t \sim \mathcal{N}(0, I), \quad (19)$$

212 for SGLD [1], and

$$v_{t+1} = \beta v_t + \eta \nabla \log p(x_t) + \sqrt{2\alpha\eta} \xi_t, \quad x_{t+1} = x_t + v_{t+1}, \quad (20)$$

213 for an SGHMC-style baseline [2]. Here η is the step size, α is a friction or
 214 noise parameter, and $\beta \in (0, 1)$ is the momentum decay.

215 *Particle configuration and kernel choice.* For particle methods, we use $N =$
 216 128 particles; for single-chain methods, we compute metrics from a rolling
 217 window of recent iterates to obtain comparable finite-sample summaries.
 218 Unless stated otherwise, particle methods use a Gaussian RBF kernel with
 219 bandwidth selected by the median heuristic and a benchmark-specific scal-
 220 ing factor. For Mix8, we instead use a shared multiscale RBF kernel with
 221 bandwidth factors $\{0.5, 1, 2\}$ to better probe local spacing and longer-range
 222 interactions across modes.

223 *Implementation safeguards.* To improve numerical robustness, the multirate
224 implementations clip raw score components to the interval $[-50, 50]$ before
225 assembling the update. Benchmarks with pronounced instability additionally
226 use a short non-finite fail-fast guard, which we note locally when it is active.

227 *Checkpointing and model selection.* Unless noted otherwise, experiments run
228 for up to 1,000 iterations over five seeds with checkpointed evaluation ev-
229 ery 20 iterations. All methods are implemented in a shared JAX codebase
230 with unified benchmarking, checkpointing, metric computation, and plotting
231 pipelines to support fair comparison and reproducibility. All benchmarks use
232 a patience-based early-stopping rule triggered by persistent degradation of
233 a designated monitor metric. Pure sampling benchmarks use KSD as the
234 monitor (50D Gaussian, 2D synthetic targets, and Mixture2D), whereas pre-
235 dictive benchmarks use held-out NLL (UCI logistic regression, BNN, and
236 HLR). For each run, we retain the best finite checkpoint under that monitor
237 before stopping. Accuracy and ECE are reported for predictive tasks but are
238 never used as stopping criteria.

239 4. Results

240 In the following, we introduce each benchmark, explain its relevance to
241 particle-based sampling, and report the corresponding empirical results. The
242 benchmark suite consists of a 50D Gaussian target, 2D synthetic targets, a
243 multimodal Gaussian mixture (Mix8), UCI logistic regression, a Bayesian
244 neural network, and large-scale hierarchical logistic regression.

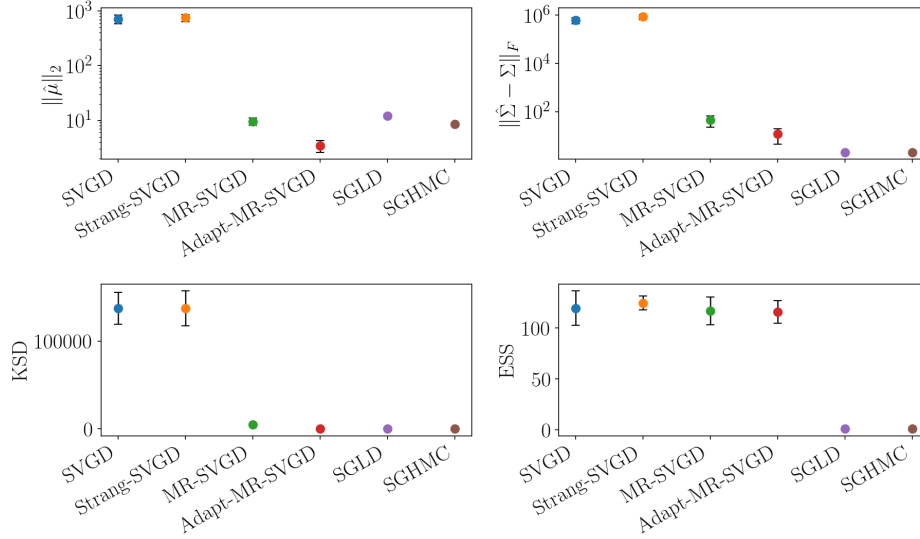


Figure 1: 50D Gaussian final-checkpoint summary across methods. The four panels report $\|\hat{\mu} - \mu\|_2$, $\|\hat{\Sigma} - \Sigma\|_F$, KSD, and ESS, respectively. Markers show mean values across seeds and error bars indicate one standard deviation. Lower is better for the first three metrics, whereas higher is better for ESS. This figure highlights robustness differences under strong anisotropy: Adapt-MR-SVGD is the only particle variant that keeps both moment errors and KSD under control.

245 4.1. 50D Gaussian

246 This benchmark tests the stability of sampling in a high-dimensional set-
 247 ting with anisotropic covariance structure. We run 1,000 sampling iterations
 248 with checkpointing every 50 steps and a 5-seed ablation. Reported met-
 249 rics are moment errors ($\|\hat{\mu} - \mu\|_2$ and $\|\hat{\Sigma} - \Sigma\|_F$), ESS, KSD, and time-to
 250 best-checkpoint.

251 Among the methods, fig. 1 shows that Adapt-MR-SVGD is the only vari-
 252 ant that keeps both moment errors and KSD under control in this anisotropic
 253 Gaussian setting: it substantially improves on MR-SVGD and avoids the se-

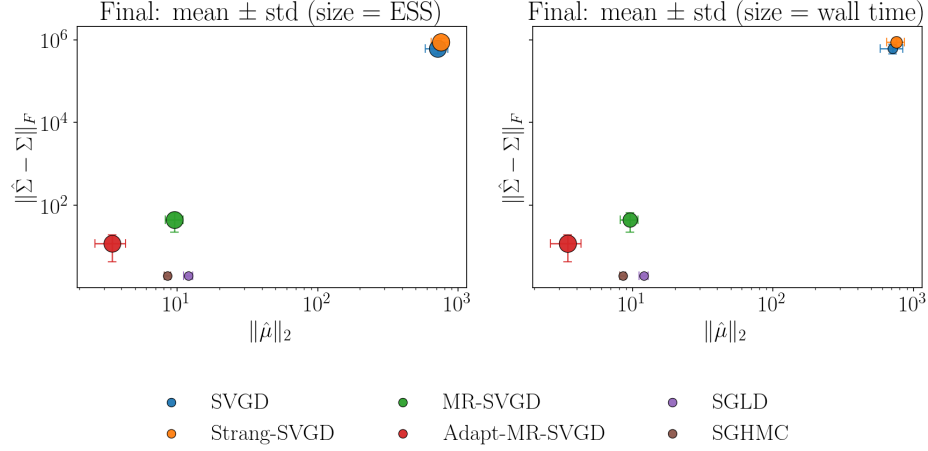


Figure 2: 50D Gaussian: final mean $\pm$ std Pareto plot in moment-error space with marker size encoding ESS (left) and wall time (right).

vere degradation exhibited by vanilla SVGD and Strang-SVGD. The chain
baselines remain competitive on some final-checkpoint fidelity metrics, but
they do so with ESS near 4, far below the particle methods. Within the
particle family, fig. 2 shows that MR-SVGD offers a cheaper but less accu-
rate baseline, whereas Adapt-MR-SVGD achieves the strongest quality at a
higher computational cost.

4.2. 2D synthetic targets

In this benchmark, we evaluate a suite of 2D targets with varying ge-
ometry and multimodality, including banana, ring, squiggly, two-moons, and
funnel. To contextualize the geometry of these targets, fig. 3 shows short-run
visualization panels for representative cases. Here KSD serves both as the
stop monitor and the primary quality metric. We add a safety guard against
non-finite KSD values in the stopping policy, because several targets can
produce unstable particle trajectories. The main comparison in table 1 fo-

268 cuses on KSD, mean log probability, gradient-evaluation cost, and wall time.
 269 The panels in fig. 3 are qualitative only and are included to illustrate target
 270 geometry and short-run particle behavior.

271 Table 1 shows a sumamry of the metrics for the 2D suite. Banana is
 272 effectively a tie: all four particle methods reach nearly identical best KSD
 273 values, with only minor differences in mean log probability and cost. The
 274 other targets separate more sharply. On ring, squiggly, and two moons,
 275 Adapt-MR-SVGD is the only method that stays near the target while keeping
 276 KSD below 1, whereas the other particle methods remain orders of magnitude
 277 worse in both KSD and mean log probability. MR-SVGD is often cheaper
 278 in gradient-evaluation count on those targets, but it pays for that reduction
 279 with substantially weaker fidelity.

280 Funnel is the main exception. There MR-SVGD attains the best median
 281 KSD, but Adapt-MR-SVGD has much better median mean log probability
 282 while also using fewer gradient evaluations and less wall time. Taken to-
 283 gether, these results show that adaptive multirate splitting is not uniformly
 284 best on every benign 2D target, but it is the only particle variant that com-
 285 bines strong distributional fidelity with competitive cost across the harder
 286 geometries.

287 4.3. *Mixture2D (mix8)*

288 Mix8 is an equal-weight eight-component Gaussian mixture with modes
 289 arranged uniformly on a ring of radius 4. All methods start from the same
 290 small Gaussian centered at the origin, away from those modes. For this
 291 benchmark we use a shared multiscale RBF kernel with bandwidth factors
 292 $\{0.5, 1, 2\}$ for all particle methods so that local spacing and longer-range

293 repulsion are both represented. This also demonstrates that the multirate
 294 formulation is agnostic to kernel choice: here we use a richer kernel than a
 295 single-bandwidth RBF to capture both local and longer-range interactions.
 296 Prior SVGD work has also considered broader kernel families beyond a single
 297 scalar RBF [5, 21].

298 Let $a(x_i)$ denote the nearest mixture center for particle x_i , and let

$$p_k = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{a(x_i) = k\}$$

299 be the empirical mass of mode k . We define mode coverage as the fraction of
 300 modes with $p_k \geq 0.05$, normalized mode entropy as $-\sum_{k=1}^K p_k \log p_k / \log K$,
 301 and mode imbalance as $\text{std}(p_1, \dots, p_K)$. Thus, higher coverage and entropy
 302 indicate broader exploration, while lower imbalance indicates more even mass
 303 allocation across discovered modes. The primary metrics are therefore mode
 304 coverage, normalized mode entropy, and mode imbalance. KSD and mean log
 305 probability are retained as secondary diagnostics. The fixed-budget results
 306 are summarized in fig. 4, which reports the mean final metrics across seeds
 307 together with one standard deviation.

308 Figure 4 shows that Adapt-MR-SVGd gives the strongest fixed-budget
 309 multimodal result. It attains the highest mean final coverage and entropy
 310 across seeds (0.500 and 0.544, respectively) while keeping KSD at 0.360, far
 311 below the single-rate particle baselines. Vanilla SVGD and Strang-SVGd
 312 spread farther around the ring than MR-SVGd, but that extra spread is
 313 unstable, with final KSD in the 10^3 range. Fixed MR-SVGd is much more
 314 stable than those baselines, yet it still underexplores the ring, reaching only
 315 0.250 coverage and 0.238 entropy.

316 The imbalance panel further shows that Adapt-MR-SVGD distributes
 317 mass more evenly across the modes it discovers. By contrast, SGLD and
 318 SGHMC remain near a single mode, with coverage fixed at 0.125 and zero
 319 entropy, so their small KSD values do not indicate successful multimodal
 320 exploration.

321 4.4. UCI logistic regression

322 We benchmark Bayesian logistic regression on four UCI datasets [22]:
 323 breast cancer, ionosphere, spambase, and a5a. All four are binary tasks
 324 with a Gaussian prior on the regression weights. We emphasize test NLL,
 325 accuracy, and ECE, with ESS reported as a supplemental mixing diagnostic.
 326 Summaries are shown in fig. 5, and the particle-method best-NLL summary
 327 is reported in table 2. In both we use the best finite-NLL checkpoint selected
 328 by the predictive stopping rule.

329 The particle-only summary in table 2 shows that Adapt-MR-SVGD at-
 330 tains the best NLL on breast cancer, ionosphere, and a5a, whereas vanilla
 331 SVGD is best on spambase. The broader comparison in fig. 5 extends this
 332 view to the chain baselines and shows that SGHMC on spambase and a5a,
 333 and SGLD on ionosphere, can achieve lower NLL at substantially lower wall
 334 time, albeit with ESS far below the particle methods. Accuracy varies little
 335 across methods on these datasets, so NLL provides the more discriminative
 336 predictive comparison. Overall, UCI logistic regression is a comparatively
 337 mild benchmark for multirate splitting: adaptive multirate updates remain
 338 competitive, but the clearest multirate gains in this paper appear on the
 339 more anisotropic, multimodal, and hierarchical targets.

340 4.5. Bayesian neural network

341 We evaluate a one-hidden-layer Bayesian neural network (width 32) on
342 Spambase and a5a. We use a Gaussian prior with one shared validation
343 split per dataset. Checkpoints are selected by validation NLL under the
344 global predictive stopping rule, and we report test NLL, accuracy, ECE, ESS,
345 and mean log probability at the selected checkpoint. Per-dataset summaries
346 appear in fig. 6.

347 The per-dataset summaries in fig. 6 show a split outcome between the
348 two datasets. On Spambase, Adapt-MR-SVGD attains the best test NLL
349 and the highest mean test accuracy, while the remaining particle methods
350 cluster tightly together. On a5a, the ranking shifts: MR-SVGD attains the
351 best overall test NLL and the highest mean test accuracy, with Adapt-MR-
352 SVGD remaining the second-best particle method and the single-rate particle
353 variants again worse. The same panels show that SGHMC on Spambase
354 and SGLD on a5a remain competitive, but with ESS far below the particle
355 methods.

356 4.6. Large-scale hierarchical logistic regression

357 To stress-test multirate behavior in a high-dimensional anisotropic poste-
358 rior, we study a large-scale hierarchical logistic regression (HLR) benchmark
359 with grouped random effects. The combination of many group-level effects
360 and a global scale parameter induces funnel-like geometry and strong local
361 stiffness, making this benchmark a demanding joint test of predictive quality,
362 robustness, and cost for the SVGD variants.

363 *Problem definition.* Given binary outcomes $y_i \in \{0, 1\}$, sparse features $x_i \in$
 364 $\mathbb{R}^p$, and group indices $g_i \in \{1, \dots, G\}$, we consider

$$\begin{aligned} y_i \mid x_i, g_i, \beta, \alpha, u &\sim \text{Bernoulli}(\sigma(\alpha + x_i^\top \beta + u_{g_i})), \\ i &= 1, \dots, n, \end{aligned} \quad (21)$$

365 where $\sigma(\cdot)$ is the logistic link. We use a non-centered parameterization for
 366 group effects,

$$\begin{aligned} \beta &\sim \mathcal{N}(0, \sigma_\beta^2 I_p), \quad \alpha \sim \mathcal{N}(0, \sigma_\alpha^2), \\ u_j &= \tau z_j, \quad z_j \sim \mathcal{N}(0, 1), \\ \log \tau &\sim \mathcal{N}(\mu_\tau, \sigma_\tau^2), \end{aligned} \quad (22)$$

367 for $j = 1, \dots, G$. The posterior (up to normalization) is

$$\begin{aligned} \pi(\theta) &\propto \left[\prod_{i=1}^n \text{Bernoulli}(y_i; \sigma(\alpha + x_i^\top \beta + u_{g_i})) \right] \\ &\quad \times p(\beta) p(\alpha) p(z) p(\log \tau), \\ \theta &= (\beta, \alpha, z, \log \tau). \end{aligned} \quad (23)$$

368 *Protocol.* Our primary HLR benchmark uses a long-tail group-frequency dis-
 369 tribution with $n = 10^6$ observations, $p = 300$ sparse features, and $G = 50,000$
 370 groups. Particle methods use 32 particles, whereas single-chain baselines are
 371 evaluated with rolling-window checkpoint summaries. As in the other pre-
 372 dictive benchmarks, checkpoint selection and early stopping are driven by
 373 predictive NLL; here we additionally stop after two consecutive non-finite
 374 checkpoints to prevent unstable runs from drifting far outside the finite-NLL
 375 regime. A uniform-group variant is reported below as a sensitivity check.

376 *Result summary.* table 3 shows the clearest predictive separation among the
 377 particle methods in the paper. Adapt-MR-SVGD attains the lowest mean

best-NLL (0.613) while remaining finite on all five seeds. MR-SVGD is the closest competitor, with only slightly worse NLL (0.623) and a lower wall time at the selected checkpoint (92.5 s versus 123.3 s). By contrast, vanilla SVGD, Strang-SVGD, and SGHMC frequently fail to reach stable finite checkpoints in this regime, while SGLD remains finite more often but with substantially worse predictive performance. In the long-tail setting, the main distinction is therefore not only final NLL, but whether the method remains numerically viable under the hierarchical funnel geometry.

Uniform-group sensitivity check. table 4 confirms the same qualitative ordering under a less pathological grouping structure. Adapt-MR-SVGD again attains the lowest mean best-NLL with full finite-seed coverage, while MR-SVGD remains close in NLL and is the fastest among methods with full finite coverage. The long-tail regime remains the more discriminative benchmark, but the relative ranking of the multirate variants is stable across both group-assignment patterns.

5. Conclusions

This work studies a central numerical weakness of SVGD: a single global step size must simultaneously resolve attractive drift and diversity-preserving repulsion even when these components evolve on different effective time scales. We address this mismatch by recasting SVGD as a split interacting-particle flow and integrating the drift and repulsion terms with multirate updates.

Across a broad benchmark suite, the main empirical pattern is that multi-rate SVGD improves robustness and quality-cost tradeoffs relative to vanilla

402 SVGD, with the strongest gains on anisotropic, multimodal, and hierarchi-
403 cal targets. Fixed multirate schedules provide a simpler robustness baseline,
404 while adaptive drift resolution offers the strongest performance when local
405 stiffness changes along the trajectory. At the same time, no single vari-
406 ant dominates every metric and every regime, reinforcing the importance of
407 evaluating quality and cost jointly rather than through a single summary
408 statistic.

409 Several limitations remain. Current adaptivity is centered on drift reso-
410 lution, while repulsion frequency is still controlled by simple schedules. ESS
411 remains a univariate proxy, and dense kernel interactions continue to con-
412 strain scaling at large particle counts. These observations suggest clear
413 next steps: tighter adaptive controllers that coordinate drift and repul-
414 sion updates, approximate or structured kernel computations for large- N
415 regimes, and stronger theoretical links between multirate stability analysis
416 and particle-based Bayesian inference guarantees. Together, these directions
417 can move multirate SVGD from a robust empirical strategy toward a more
418 principled and scalable foundation for modern Bayesian computation.

419 **Data availability**

420 Code, experiment scripts, and manuscript sources for this study are main-
421 tained in a public GitHub repository.² Public benchmark inputs used in the
422 predictive experiments are derived from standard UCI datasets. All other
423 benchmark targets are synthetic and are generated directly by the repository

²<https://github.com/csml-beach/multirate-sampling>

code. Generated figures, metrics, and supplementary animation artifacts are available in the project repository for reference.

Declaration of generative AI and AI-assisted technologies

The JAX codebase for this project was developed in part with the assistance of OpenAI Codex. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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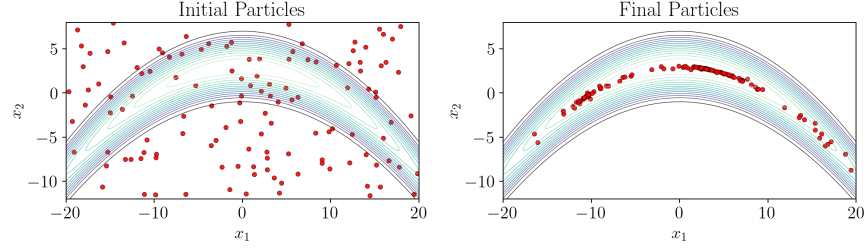
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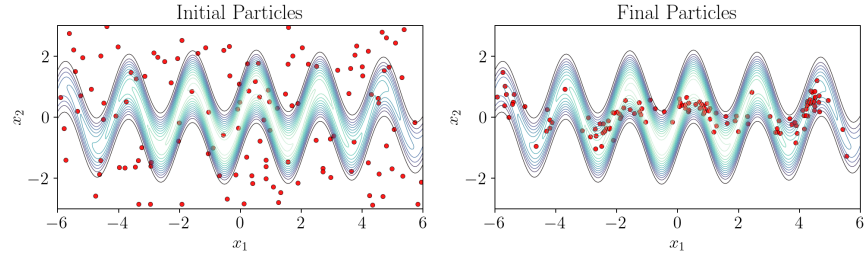
Algorithm 2 Adaptive multirate SVGD (Adapt-MR-SVGD)

Require: Particles $\{x_i\}_{i=1}^N$, step size h , bounds $m_{\min}, m_{\max}$, tolerance Tol

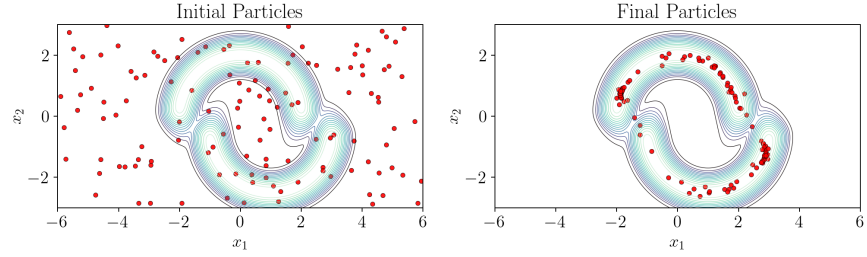
```
1:  $x \leftarrow \{x_i\}$ 
2: compute repulsion  $f_{\text{rep}}(x)$  using eq. (5)
3:  $x \leftarrow x + h f_{\text{rep}}(x)$ 
4: compute drift  $f_{\text{drift}}(x)$  using eq. (4)
5:  $x_{\text{full}} \leftarrow x + h f_{\text{drift}}(x)$ 
6:  $x_{\text{half}} \leftarrow x + (h/2) f_{\text{drift}}(x)$ 
7: compute drift  $f_{\text{drift}}(x_{\text{half}})$ 
8:  $\tilde{x}_{\text{full}} \leftarrow x_{\text{half}} + (h/2) f_{\text{drift}}(x_{\text{half}})$ 
9: estimate relative error  $\rho = \|\tilde{x}_{\text{full}} - x_{\text{full}}\|/(\|x\| + \epsilon)$ 
10:  $m \leftarrow \text{clip}(\lceil \sqrt{\rho/\text{Tol}} \rceil, m_{\min}, m_{\max})$ 
11: if  $m \leq 2$  then
12:    $x \leftarrow \tilde{x}_{\text{full}}$   $\triangleright$  reuse current predictor if accurate
13: else
14:   for  $s = 1$  to  $m$  do
15:     compute drift  $f_{\text{drift}}(x)$ 
16:      $x \leftarrow x + (h/m) f_{\text{drift}}(x)$ 
17:   end for
18: end if
19: return  $x$ 
```



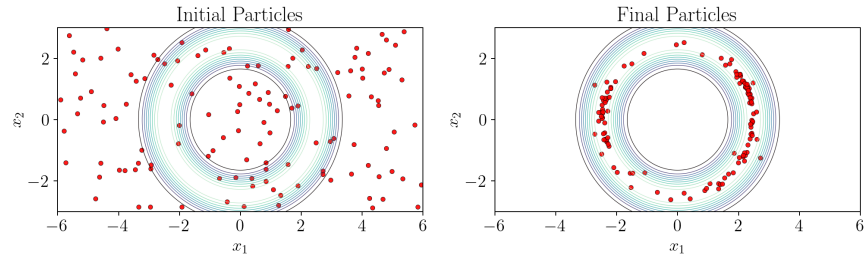
(a) Banana.



(b) Squiggly.



(c) Two moons.



(d) Ring.

Figure 3: 2D target visualization panels produced with Adapt-MR-SVGD under visualization-only settings. Each subfigure shows target-density contours with initial particles (left) and short-run final particles (right).

Method	Best KSD↓	mean logp@best↑	grad@best↓	wall@best (s)↓
<i>Banana</i>				
Strang-SVGD	0.114	-0.426	100	1.35
MR-SVGD	0.114	-0.392	80	1.13
Adapt-MR-SVGD	0.114	-0.393	160	0.80
SVGD	0.114	-0.391	80	0.32
<i>Funnel</i>				
MR-SVGD	0.256	-3.136	160	1.24
Adapt-MR-SVGD	0.357	-0.197	84	0.40
Strang-SVGD	0.369	-9.289	120	1.46
SVGD	0.377	-6.801	160	0.46
<i>Ring</i>				
Adapt-MR-SVGD	0.226	-0.004	238	0.82
MR-SVGD	64.35	-815.6	50	0.70
SVGD	131.6	-3753	20	0.16
Strang-SVGD	665.7	-7.92×10^5	240	2.79
<i>Squiggly</i>				
Adapt-MR-SVGD	0.719	-0.910	826	2.02
Strang-SVGD	22.49	-5.25×10^4	200	2.47
SVGD	22.60	-2.81×10^4	80	0.32
MR-SVGD	54.04	-422.7	100	1.36
<i>Two moons</i>				
Adapt-MR-SVGD	0.315	-0.009	182	0.67
MR-SVGD	134.3	-1250	40	0.60
Strang-SVGD	1892	-1.84×10^6	120	1.55
SVGD	2085	-2.58×10^6	80	0.34

Table 1: 2D particle-method summary over five seeds. For each target and method, we report medians over seeds at the best finite-KSD checkpoint. Methods are ordered by median KSD within each target block. Banana shows little separation, funnel remains nuanced, and Adapt-MR-SVGD is strongest on ring, squiggly, and two moons.

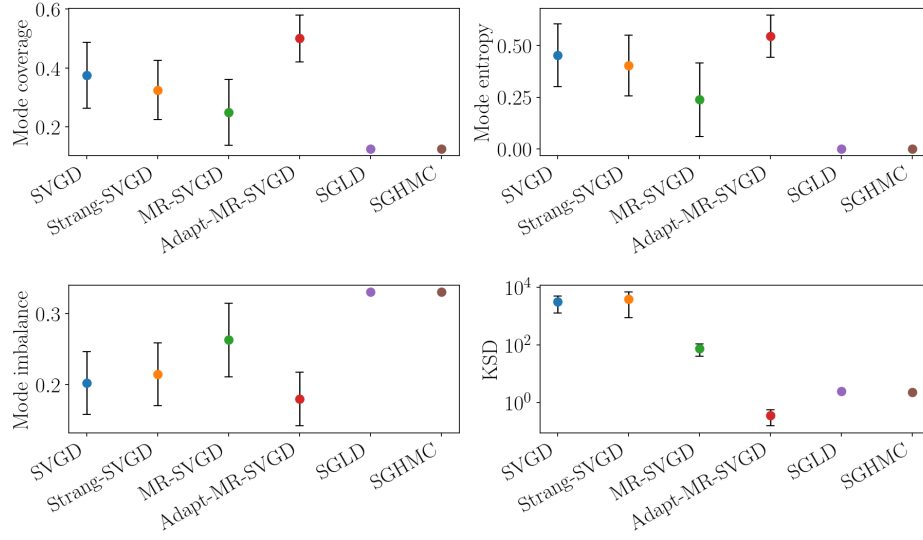
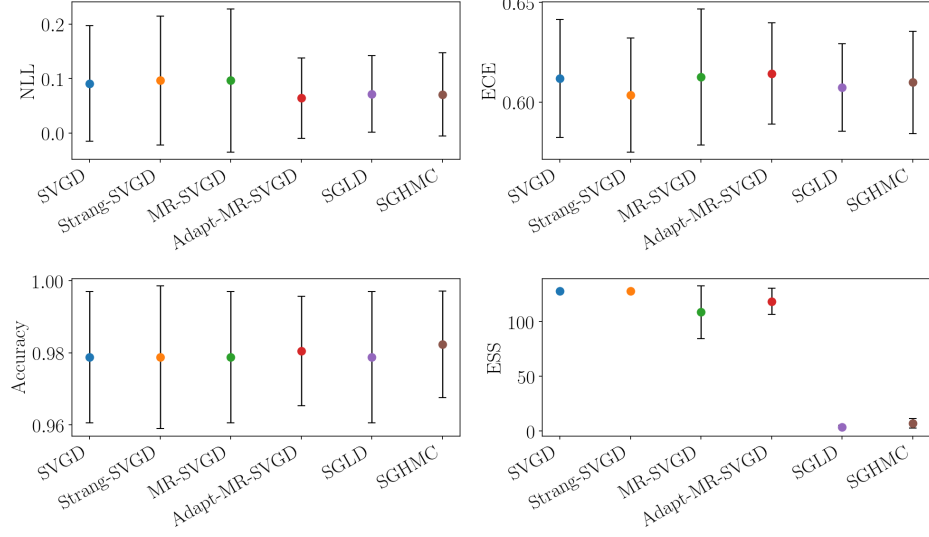
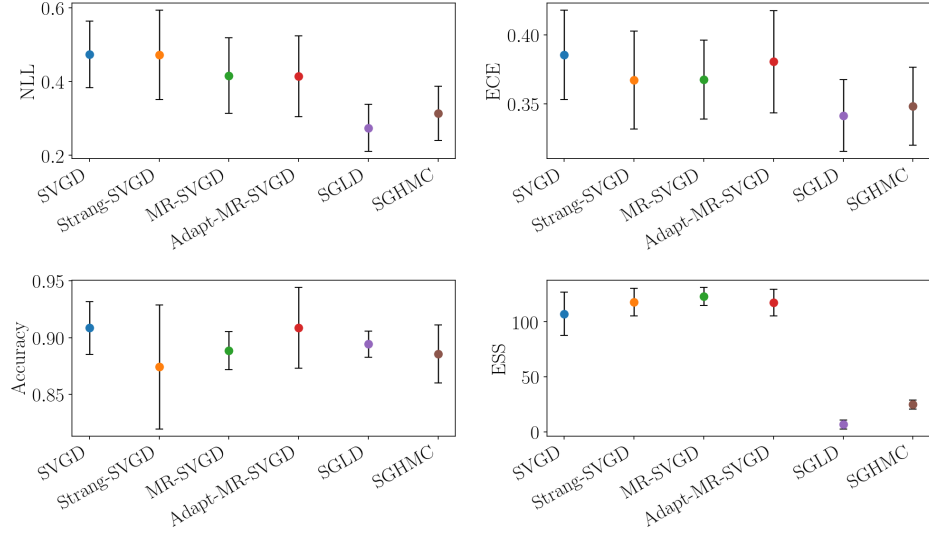


Figure 4: Mixture2D (mix8) fixed-budget final-checkpoint summary across methods. The four panels report mode coverage, mode entropy, mode imbalance, and KSD. Higher is better for coverage and entropy, while lower is better for mode imbalance and KSD.

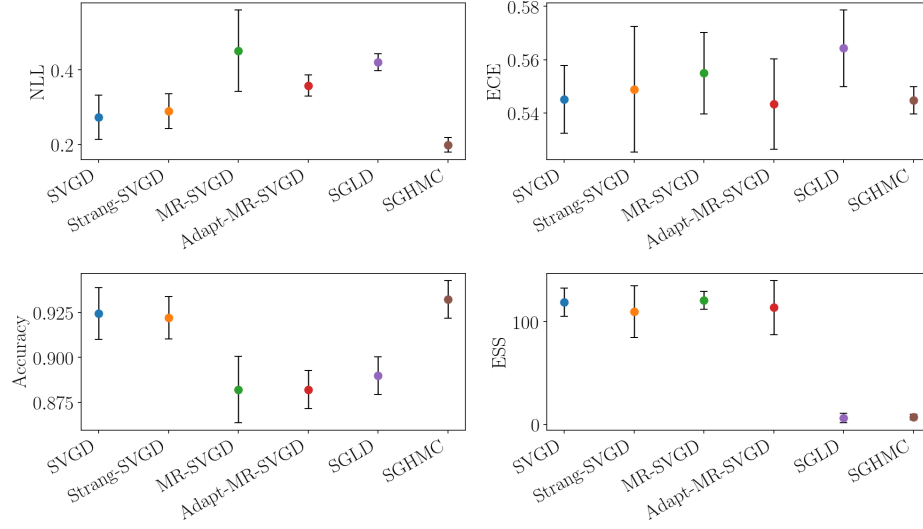


(a) Breast cancer

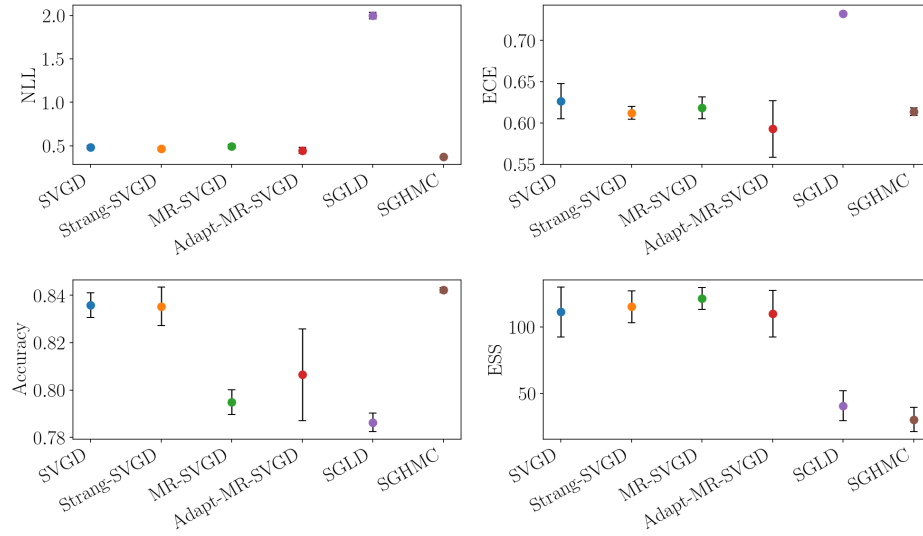


(b) Ionosphere

Figure 5: UCI logistic regression summary across datasets. Each subpanel reports test accuracy, NLL, ECE, and ESS at the best finite-NLL checkpoint selected by the predictive stopping rule. Remaining datasets are shown in the continued panel.



(c) Spambase

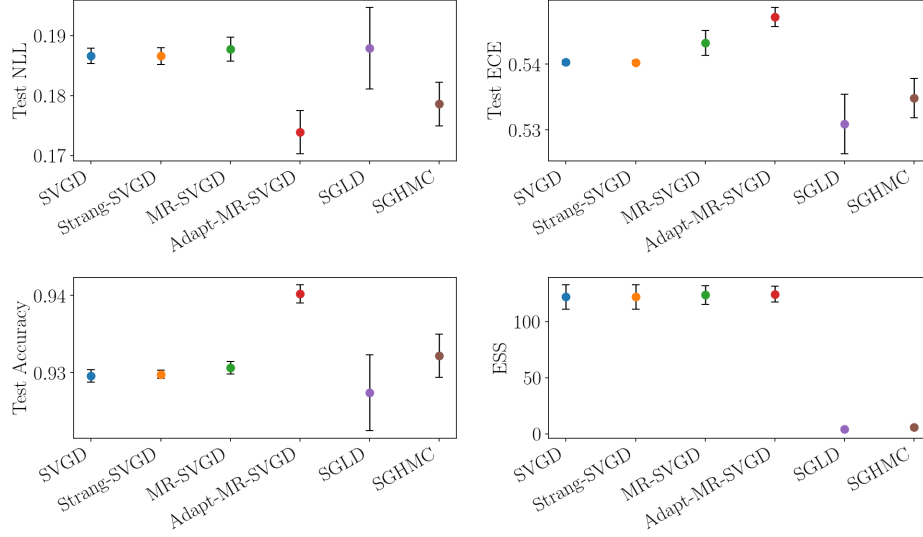


(d) a5a

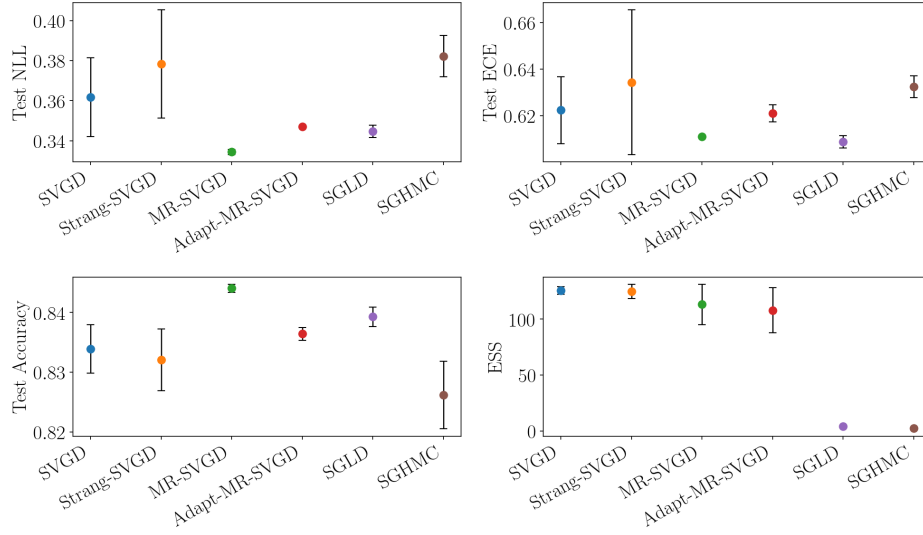
Figure 5: UCI logistic regression summary across datasets (continued).

Method	Best NLL↓	Acc@best NLL↑	Wall@best (s)↓
<i>Breast cancer</i>			
Adapt-MR-SVGD	0.028	0.987	3.7
SVGD	0.091	0.979	1.4
Strang-SVGD	0.096	0.979	4.0
MR-SVGD	0.096	0.979	3.4
<i>Ionosphere</i>			
Adapt-MR-SVGD	0.414	0.909	5.2
MR-SVGD	0.416	0.889	3.9
Strang-SVGD	0.473	0.874	3.0
SVGD	0.473	0.909	0.6
<i>Spambase</i>			
SVGD	0.273	0.925	0.8
Strang-SVGD	0.289	0.922	3.3
Adapt-MR-SVGD	0.358	0.882	6.6
MR-SVGD	0.452	0.882	3.9
<i>a5a</i>			
Adapt-MR-SVGD	0.444	0.806	14.2
Strang-SVGD	0.463	0.835	2.9
SVGD	0.482	0.836	1.6
MR-SVGD	0.500	0.793	6.6

Table 2: UCI logistic-regression particle-method summary. For each dataset and method, we average over the finite best-NLL checkpoints selected per seed and report NLL, the corresponding accuracy, and wall time. ECE is summarized in the per-dataset panels of fig. 5.



(a) Spambase



(b) a5a

Figure 6: BNN predictive summary across datasets. Each subpanel reports test accuracy, NLL, ECE, and ESS at the checkpoint selected by validation NLL.

Method	Finite seeds $\uparrow$	Best NLL $\downarrow$	Acc@best NLL $\uparrow$	Wall@best (s) $\downarrow$
Adapt-MR-SVGD	5/5	0.613	0.658	123.3
MR-SVGD	5/5	0.623	0.656	92.5
SGHMC	2/5	0.860	0.560	77.9
SVGD	3/5	0.885	0.577	154.2
Strang-SVGD	3/5	1.263	0.579	163.6
SGLD	4/5	1.823	0.576	101.6

Table 3: HLR long-tail summary over five seeds. For each seed and method, we select the best finite-NLL checkpoint and report method-wise means of that NLL, the accuracy at the selected checkpoint, and the corresponding wall time. “Finite seeds” counts runs that reached at least one finite-NLL checkpoint before stopping. This is the main hierarchical stress test: Adapt-MR-SVGD attains the lowest mean best-NLL while remaining finite on all seeds, and MR-SVGD is the closest lower-cost baseline.

Method	Finite seeds $\uparrow$	Best NLL $\downarrow$	Acc@best NLL $\uparrow$	Wall@best (s) $\downarrow$
Adapt-MR-SVGD	5/5	0.681	0.601	94.6
SVGD	1/5	0.700	0.548	54.9
MR-SVGD	5/5	0.706	0.597	44.2
Strang-SVGD	1/5	0.708	0.584	241.7
SGLD	5/5	0.743	0.528	66.0
SGHMC	5/5	0.791	0.554	79.7

Table 4: HLR uniform-group summary over five seeds. Entries are defined as in table 3, but with balanced group assignments as a sensitivity check.